Dual Image Super-Resolution for High-Resolution Optical Satellite Imagery and Its Blind Evaluation Using an Agentic Approach

***Abstract*—The demand for high resolution satellite images is increasing, but its gathering is constrained by sensor, bandwidth, and operational cost. Super-resolution (SR) provides a computational solution by predicting HR images from LR images, but most existing deep neural network models are static and may hallucinate details, making them less reliable in science or defence applications. A coordinated Dual Image Super-Resolution (DISR) framework based on Agentic AI-guided blind evaluation and self-correction pipeline is proposed in this paper. It applies U-Net, a VGG16-based CNN, and a Custom CNN to the WorldStrat dataset2 with paired Sentinel-2 (10 m) and Airbus SPOT (1.5 m) archival images to enhance the images. The framework consists of dual-image preprocessing, model training, agentic model selection, no-reference quality assessment, and automated correction through an unlearning controller. Experimental results demonstrate that the proposed scheme yields 32.7 dB PSNR and 0.85 SSIM, which is far better than the static CNN based and GAN based methods that merely obtains 19.23 dB PSNR and 0.58 SSIM. The approach is an adaptive and end-to end satellite image enhancement method with less manual effort.**

***Index Terms*—Dual Image Super Resolution, Agentic AI selector, Blind Evaluation, U Net, VGG16 CNN, Custom CNN, PSNR, SSIM, MAE**

I. Introduction

In the age of digitalization, high resolution (HR) satellite images have been widely used in updating and especially as a data source in urban planning, environment monitoring, defence surveillance, disaster management, precision agriculture, and GIS analysis [1]. The Inspection of the Earth at higher spatial resolution reveals more information on natural and anthropogenic processes, but HR images are still difficult to obtain because of cost, hardware limitations, and data acquisition restriction from orbital altitude and sensor capacity.

Satellite sensors operate at different spatial resolutions. Sentinel-2 delivers 10-meter imagery, while commercial satellites such as Airbus SPOT or WorldView offer sub-2 meter data. This gap between LR and HR imagery restricts applications needing detailed visual information. Direct HR acquisition requires advanced sensors, expensive hardware, and restrictive licensing, making it impractical in many cases. Super-resolution (SR) has therefore become a practical method to obtain HR imagery from LR inputs.

Image Super-Resolution (SR) recovers HR details from LR observations. Deep learning-based SR methods using CNNs, GANs, and transformers have shown strong LR-HR mapping capability and improved visual quality. However, most models rely on synthetic degradations created with known downsampling kernels and do not capture real effects such as atmospheric distortion, sensor noise, or motion blur. This leads to hallucinated textures or artifacts that reduce reliability for scientific or defence use.

A further limitation is that SR models remain static. Once trained, they cannot detect errors or adapt to unseen degradations, and performance degrades on inputs affected by clouds, shadows, or illumination variation. Addressing these cases usually requires human intervention, retraining, and fine-tuning, increasing computation and effort. Hence, an expectation is emerging for autonomous, adaptive SR systems with blind quality assessment.

We propose a Dual Image Super Resolution (DISR) framework and an Agentic AI-based blind evaluation model for intrinsic validation. The framework aggregates multiple LR images of the same area, captured at different times or under different weather conditions, to produce a single HR output with enhanced spatial and spectral details. An agentic loop performs self assessment and self correction using no-reference image quality assessment (IQA) metrics, and an unlearning controller updates model weights when low-quality results are detected.

Blind evaluation is crucial because PSNR and SSIM require reference HR images, which are often unavailable in real satellite scenarios. Unlike blind IQA methods such as NIQE or PIQ that estimate perceptual quality, our approach avoids explicit perceptual prediction, improving robustness for practical settings.

This study uses the WorldStrat Dataset from Kaggle, containing co-registered Sentinel-2 (10 m) and Airbus SPOT (1.5 m) image pairs across diverse terrains. Multiple SR models, including U-Net, VGG16-based CNN, and a custom PyTorch model, are unified under a selection strategy that dynamically chooses the most suitable model based on scene characteristics such as brightness, contrast, and cloud coverage.

II. Literature Survey

Super-resolution plays an essential role in enhancing satellite imagery by generating HR images from LR inputs. Traditional SISR methods rely on deterministic mappings that fail under real-world degradations found in satellite data, often producing blurred outputs with hallucinated textures and limited generalization. Deep learning-based SR using CNNs, GANs, and transformer architectures improves fine-grained

spatial recovery but still depends on synthetic degradation models that do not represent complex and unknown degradations in actual satellite images.

Blind Super-Resolution (BSR) has been proposed to handle unknown degradations, and emerging techniques such as Agentic AI, blind IQA, and unlearning provide promising directions for autonomous and self-improving SR systems. Fang et al. [1] proposed a CNN-based NR IQA for SR using patch-wise processing and MSE optimization to subjective MOS, achieving MSE 1.2527, PLCC 0.9156, and SRCC 0.83, outperforming earlier IQA metrics.

Karwowska et al. [2] surveyed SR for small-satellite imagery across interpolation, sharpening, multi-image fusion, and deep learning, noting that GAN-based models enhance perceptual quality but may reduce PSNR. They emphasized lightweight architectures and multi-metric evaluation for computationally constrained scenarios.

Sharma et al. [3] described CNN structures including residual, recursive, GAN, and attention layers and evaluated various upsampling and loss functions, outlining efficiency–fidelity trade-offs and recommending lightweight designs for blind and dual-image SR tasks.

Cao et al. [4] proposed PCNet, a no-reference SR paradigm that estimates degradation kernels and aligns LR inputs using a triadic kernel estimation module, Fourier-domain correction, and refinement network. PCNet achieved PSNR 36.74 dB and SSIM 0.949 on Set5, surpassing KernelGAN and ZSSR while enabling fast modular blind evaluation suitable for satellite applications.

Benecki et al. [5] introduced a validation framework for multi-image SR using Sentinel-2, SPOT-5, and WorldView datasets, evaluating 17 metrics and showing that KFSSIFT and PSNRLS in GA-FRSR improved texture sharpness and structural fidelity without reference images.

Liu et al. [6] developed DLGNN, a dual-learning GNN that constructs cross-scale similarity graphs and enforces SR→LR consistency, achieving PSNR 26.45–27.12 dB on 3K VEHICLE SR and Massachusetts Roads datasets, outperforming IGNN, RCAN, and SAN for large-scale satellite processing.

Liu et al. [7] reviewed blind SISR methods and proposed a taxonomy based on degradation modeling and data sources while identifying gaps in GAN stability and implicit modeling.

Dixit et al. [8] reviewed CNN-based SISR, comparing classical and modern approaches, analysing upsampling schemes, and outlining trade-offs between speed and quality across architectures such as FSRCNN, RCAN, and ESRGAN.

Greza et al. [9] proposed a versatile SR network (VSISR) addressing high-frequency detail and radiometric consistency in satellite imagery. Using a reference HR image during inference, termed Dual Image Super Resolution, the approach reduces hallucinations while preserving radiometric characteristics. With an SRGAN-based architecture and mixed pixel augmentation, trained on Landsat, Sentinel-2, PlanetScope, and SPOT-6, VSISR achieved PSNR 25.30 dB and SSIM 0.8098, performing well on urban regions.

Ibrahim et al. [10] introduced SWViT-RRDB combining Swin Transformer attention with Residual-in-Residual Dense Blocks for satellite SR. Residual fusion groups with RRDB and overlapping cross-attention blocks enhance global and local features. Trained on OLI2MSI and Alsat-2B, SWViT-RRDB-L achieved PSNR 38.48 dB, SSIM 0.9557 on OLI2MSI and outperformed leading methods including HAT-L for small-object recovery.

Zhang et al. [11] proposed a two-stage blind SR network using dynamic kernel estimation and structural prior with triple path attention groups. DCLS embeds degradation, and TPAG extracts structural features using channel-split and global texture fusion. Trained on DIV2K and Flickr2K with Gaussian and anisotropic kernels, the method achieved strong scores on Urban100 and Manga109 with improvements over DCLS.

Zuo et al. [12] introduced 4KAgent, an agentic SR generalist for autonomous 4K upscaling using iterative planning and quality-driven restoration. A Profile module, Perception Agent with blind IQA, and Restoration Agent with mixture-of-experts enable domain-agnostic SR across 26 benchmarks, achieving state-of-the-art fidelity and perceptual quality without retraining.

Yildirim et al. [13] proposed IKR-Net, an iterative blind SISR model jointly refining SR images and blur kernels using modules for reconstruction, kernel estimation, and noise estimation with unfolding-based optimization. Handling Gaussian and motion blur degradations with noise, IKR-Net showed strong generalization at ×2 and ×4 scales on real LR images.

Dong et al. [14] proposed MTSR-GAN for Sentinel-2 enhancement from 10 m to 2.5 m using K-GAN to estimate degradation kernels and create realistic LR-HR pairs. This enables GAN-based SR preserving spectral and spatial fidelity beyond bicubic models, improving spatial detail reconstruction for Sentinel-2 scenes.

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From the literature survey, it is clear that Dual Image Super Resolution (DISR) remains largely unexplored in satellite imaging, as most existing methods focus on single-image SR trained on synthetic degradations that do not match real data. GAN models often introduce artifacts, while CNN models struggle with fine detail recovery. Multi-image SR and dual learning ideas from Benecki et al. [5] and Liu et al. [**?**] show promise, yet blind IQA components and unlearning-based optimization remain rarely integrated into SR systems. These gaps highlight the need for dual-image fusion in real satellite datasets, agentic AI driven model selection, blind evaluation for real time quality assessment, and unlearning enabled self improvement for robust and adaptive SR.

III. Materials and Methods

A. Dataset

WorldStrat was built as the first benchmark for Earth observation learning-based super-resolution. It is based on two primary satellite platforms: the European Space Agency (ESA)'s Sentinel-2 imagery and the significantly higher spatial resolution Airbus SPOT imagery. Low-resolution (10 m/px) images were obtained from the Sentinel-2 data, while high-resolution (1.5 m/px) ground-truth was provided by the Airbus SPOT.

Fig. 1. High Resolution Image with Corresponding pair of Low Resolution Images [16]

Each image pair is provided in form of a 256×256 patch to keep the input size the same for training neural networks. The dataset spans multiple continents and a variety of landscapes so that models aren't overfit to a single geography or climate. The data have been balanced geographically so that each continent and land cover type (urban, vegetation, water body, desert) are equally represented in the dataset.

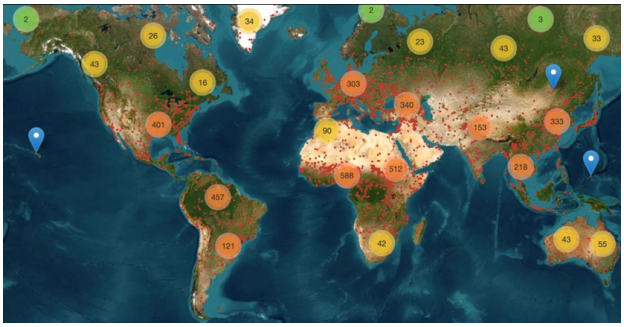

Fig. 2. Area in the Dataset [16]

B. Architecture of the System

The DISR system with agentic blind evaluation integrates deep learning, computer vision, and adaptive agent-based intelligence. Unlike single-image SR, the framework processes a pair of LR satellite images captured under different atmospheric or temporal conditions, enabling the extraction of complementary cues such as edge variation and reflectance differences. The architecture consists of Input Preprocessing, Dual-Image Fusion, SR Model Ensemble (U-Net, VGG16-based CNN, Custom CNN), Agentic Model Selector, Blind IQA Engine, and an Unlearning Controller. These operate under a closed-loop mechanism where output quality governs whether the pipeline proceeds, corrects itself, or updates internal parameters.

C. Phases of the Proposed System

Phase 1: Data Acquisition and Preprocessing. WorldStrat LR-HR pairs are resized, normalized, denoised, contrast-enhanced, cloud-filtered, and augmented, then divided into patches for efficient training. This ensures uniformity before dual-image fusion.

Phase 2: Dual-Image Fusion. Two aligned LR inputs are registered using feature-based matching and concatenated into a six-channel tensor. This fusion preserves richer spatial-temporal details and provides complementary information unavailable in single-image SR.

Phase 3: Feature Extraction and Reconstruction. The fused tensor is processed using U-Net, VGG16-based CNN, and a lightweight Custom CNN. Reconstruction is optimized through a hybrid loss combining MSE, Perceptual Loss, and Edge Loss to ensure numerical accuracy and perceptual fidelity.

Phase 4: Agentic Model Selection. The Agentic Selector evaluates scene characteristics such as brightness, entropy, and cloud ratio to select the best SR model. The BIQA module applies NIQE and PIQE for no-reference quality assessment.

Phase 5: Blind IQA Evaluation. Since reference HR data is unavailable, NIQE and PIQE provide perceptual scoring. Outputs exceeding predefined thresholds are flagged as low quality.

Phase 6: Unlearning Controller. When BIQA indicates degradation, the controller selectively updates network components, adjusts learning rates, or modifies filters responsible for poor reconstruction, enabling iterative self-correction.

Phase 7: Visualization and Output Generation. PSNR, SSIM, NIQE, and PIQE scores are logged, and visual comparisons highlight texture and contrast recovery. Outputs and metrics are stored for benchmarking and further refinement.

D. System Workflow Summary

The DISR system converts two LR inputs into an HR output through an adaptive closed-loop mechanism. Preprocessing aligns and prepares the LR pair, dual-image fusion forms an enriched tensor, and the SR ensemble reconstructs the HR image. Agentic selection and blind evaluation determine output acceptance. Low-quality results trigger the unlearning controller, while acceptable outputs are saved with metrics. Over iterations, the framework builds a self-improving performance repository.

E. Algorithmic Design Overview

The system relies on two complementary design strategies: dual-image reconstruction using deep SR networks and an agentic evaluation–unlearning mechanism for autonomous self-correction. Together, these form a hybrid architecture that blends data-driven learning with self-regulating intelligence. The system progressively improves spatial resolution while learning to detect, quantify, and correct its own errors, unlike fixed conventional SR models.

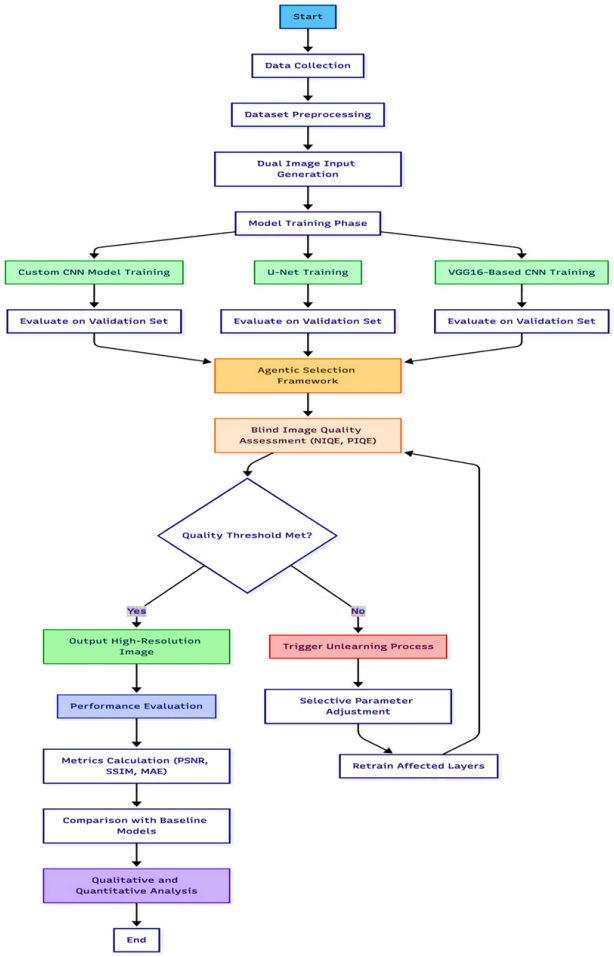

Fig. 3. Algorithm flow

F. Dual Image Super-Resolution

DISR fuses two complementary LR views of the same area to recover HR detail lost due to noise, atmospheric distortions, or illumination variance. After co-registration using feature-based matching, each LR pair is smoothed, contrast-equalized, resized, and concatenated into a six-channel tensor. This tensor is processed by U-Net, VGG16-based CNN, and a Custom CNN, each specializing in texture recovery, deep structural features, or efficient inference. Formally, if I_{LR1} and I_{LR2} denote aligned LR inputs, the network learns $I_{HR}^{pred} = f_\theta([I_{LR1}, I_{LR2}])$ and optimizes a composite loss balancing pixel fidelity, perceptual similarity, and edge sharpness. Training employs Adam optimization, ReLU activations, batch normalization, and dropout for stability and generalization. After convergence, the model generalizes to unseen LR pairs and recovers fine spatial structures more reliably than classical SISR.

G. Agentic Blind Evaluation and Unlearning

The Agentic Blind Evaluation and Unlearning (ABEU) mechanism enables the system to evaluate and refine its outputs without ground-truth access. After reconstruction, NIQE and PIQE assess naturalness and artifact levels. If scores exceed thresholds T_{NIQE} and T_{PIQE}, the Unlearning Controller selectively adjusts parameters associated with errors while preserving stable components. The update step is given by $\theta_{t+1} = \theta_t - \eta \nabla L_{corr}$, where L_{corr} emphasizes perceptual fidelity. This process repeats until scores fall within acceptable bounds. A feedback memory stores past corrections to improve future decisions. The mechanism allows the system to adapt to new imaging conditions and mitigates overfitting. Through iterative evaluation and correction, the model produces visually stable HR images across diverse LR inputs and gradually evolves toward robust, autonomous SR performance.

H. Data Preprocessing

Input images go through several preprocessing steps including resizing, cropping, noise reduction, geometric registration, normalization, augmentation, and dual-image fusion preparation. These steps standardize input size, suppress noise, and increase robustness. After preprocessing, the dataset becomes a normalized and diversified two-input sample set suitable for training deep convolutional SR models.

To ensure fair evaluation and measure performance on unseen data, the dataset is splitted into three parts: 70% for training, 15% for validation, and 15% for testing. This separation prevents the model from memorizing spatial patterns and ensures generalization to new geographical regions. Care is taken to avoid common geographic areas between training and testing to prevent over-estimated performance.

I. Computing Environment and Training Configuration

The algorithm was implemented on a high-performance computing platform to support computation-intensive deep learning–based satellite image super-resolution. All code was written in Python 3.10 on Google Colab Pro with GPU support. PyTorch 2.0 was used due to its dynamic computation graph, modular design, and extensibility for custom architectures, and its CUDA integration enables efficient matrix and convolution operations essential for SR.

The system uses an NVIDIA Tesla T4 GPU with 16 GB VRAM, an Intel Xeon 2.30 GHz processor, 16 GB RAM, and approximately 80 GB cloud storage. CUDA 12.1 was employed to parallelize GPU computations and accelerate training relative to CPU-only execution.

Training for VGG16, U-Net, and the Custom CNN used a learning rate of 1e–4, batch size of 16, and 100 epochs for stable convergence. The Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) was used for adaptive gradient updates. The loss function combined MSE, Perceptual Loss, and Edge Loss to balance pixel-level accuracy and visual realism.

This combination of GPU acceleration, optimized deep learning libraries, and cloud-based infrastructure ensures high training efficiency, reproducibility, and consistent performance, enabling the system to process large satellite datasets, handle complex neural architectures, and support iterative agentic assessment and self-improvement.

J. Metrics for Evaluation

Evaluation metrics are essential for assessing the performance of image recovery systems. Since satellite datasets do not always provide ground-truth HR references, both reference-based and no-reference metrics are used. In this work, PSNR and SSIM act as traditional reference-based measures, while NIQE and PIQE serve as blind perceptual metrics. Together, these metrics provide a complete view of quantitative fidelity and perceptual realism.

1) Peak Signal-to-Noise Ratio (PSNR): PSNR evaluates the quality of reconstructed images with respect to the ground-truth. Larger PSNR indicates higher fidelity. It is defined as

$$PSNR = 10\log_{10}\left(\frac{MAX_I^2}{MSE}\right),$$

where MAX_I is the maximum pixel value and MSE is the mean squared error. A PSNR above 30 dB generally reflects high-quality reconstruction.

2) Structural Similarity Index Measure (SSIM): SSIM quantifies perceptual degradation by comparing luminance, contrast and structure between the reconstructed and reference images. It is expressed as

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)},$$

where μ, σ^2, and σ_{xy} denote means, variances and covariance. SSIM close to 1 indicates high structural similarity.

3) Natural Image Quality Evaluator (NIQE): NIQE is a blind metric that measures deviation from natural scene statistics. Lower NIQE values indicate more natural and visually realistic images. It is a primary indicator for blind evaluation in the proposed system.

4) Perception-based Image Quality Evaluator (PIQE): PIQE estimates perceptual distortion by analysing local blocks for blur, noise and artifacts. Lower PIQE corresponds to better perceptual quality. Together with NIQE, it provides a balanced assessment of local and global degradations.

5) Mean Absolute Error (MAE): MAE measures pixel-wise accuracy between the reconstructed HR and reference HR images,

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|I_i^{pred} - I_i^{gt}|.$$

Lower MAE indicates closer pixel correspondence.

K. Parameter Settings

Model performance depends on proper parameter configuration. Each input consists of two LR satellite images I_{LR1} and I_{LR2} of the same scene. They are overlapped to form a $256 \times 256 \times 6$ tensor scaled to $[0, 1]$. The dataset is split into 70 percent training, 15 percent validation and 15 percent testing. Training uses the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, a learning rate of 1×10^{-4} decayed by 0.1 every 30 epochs, batch size 16 and 100 total epochs. Early stopping with patience 10 is employed.

The composite loss combines MSE, Perceptual Loss and Edge Loss with weights $\alpha = 0.6$, $\beta = 0.3$ and $\gamma = 0.1$. NIQE and PIQE are used as agentic blind evaluation boundaries, defined as NIQE ≤ 4.0 and PIQE ≤ 30. If these thresholds are exceeded, selective unlearning is triggered by fine-tuning specific convolutional layers with a reduced learning rate between 1×10^{-5} and 1×10^{-6}.

All experiments were conducted on Google Colab Pro with an NVIDIA Tesla T4 GPU (16 GB VRAM), Intel Xeon CPU at 2.30 GHz, 16 GB RAM and CUDA 12.1 running under Ubuntu 20.04.

IV. Results and Discussion

This section presents both quantitative and qualitative results of the proposed framework. Three baseline models, Custom CNN, U-Net and VGG16-based CNN, were trained independently on the WorldStrat dataset using dual LR pairs as input. Their outputs were then integrated into the Agentic Selection and Unlearning framework for blind evaluation and self-correction. Results are analysed first for individual models before the agentic loop, followed by the overall improvement after applying agentic evaluation, blind IQA and adaptive unlearning.

A. Baseline CNN Model

The baseline CNN contained five convolutional layers and two fully connected layers and served as a lower bound for subsequent models. It processed the dual LR inputs concatenated into a six-channel tensor and used MSE loss for pixel-wise accuracy. The baseline produced HR outputs but failed to recover high-frequency textures in urban, vegetation and mountainous areas, leading to oversmoothed reconstructions and loss of structural detail.

TABLE I
CNN performance

PSNR	SSIM	NIQE	PIQE
19.23 dB	0.58	4.71	27.4

The PSNR of 19.23 dB indicates moderate reconstruction quality, and SSIM of 0.58 shows weak structural similarity. NIQE and PIQE values reflected acceptable but not strong perceptual naturalness. Visual inspection confirmed oversmoothing, blurred vegetation and missing edges in urban areas.

B. U-Net Based Model

The U-Net improved feature reconstruction due to its encoder-decoder structure with skip connections. Context features were extracted from the LR pair and spatial details were restored during decoding. Skip connections compensated for texture loss during downsampling.

TABLE II
U-Net performance

PSNR	SSIM	NIQE	PIQE
28.70 dB	0.76	3.92	22.1

The U-Net achieved a gain of 9.47 dB in PSNR and improved SSIM to 0.76 compared to the baseline. Lower NIQE and PIQE scores indicated enhanced perceptual realism. Visually, edges were sharper, vegetation smoother and cloud regions clearer, although small artifacts persisted under motion blur or illumination mismatch.

C. VGG16-Based CNN Model

The VGG16 model further improved reconstruction by using deeper convolutional layers and pretrained feature extraction. Transfer learning enabled better generalization across diverse satellite scenes.

TABLE III
VGG16 PERFORMANCE

PSNR	SSIM	NIQE	PIQE
30.50 dB	0.81	3.65	20.8

The VGG16-based model achieved PSNR 30.5 dB and SSIM 0.81, showing strong structural preservation and perceptual fidelity. The improved NIQE (3.65) and PIQE (20.8) reflect superior naturalness compared to U-Net and the baseline. The model produced sharper reconstructions in high-contrast city regions, capturing man-made structures and edges clearly, though at higher computational cost. It remained the best-performing model before applying the agentic loop.

D. Agentic AI-Based Dual Image Super-Resolution (Proposed System)

Fig. 4. Merged SR image generated from our Agentic AI model

The Agentic Dual Image Super-Resolution model applies agentic selection on the outputs of the three base architectures. The system blindly evaluates reconstructed images using NIQE and PIQE, selects the result with the best perceptual consistency, and activates the unlearning module for targeted fine-tuning when needed. The complexity of the models increases with their performance, and the agentic system attains the highest PSNR at the cost of greater parameter count, whereas the Basic CNN and Simple U-Net remain lightweight.

The qualitative evaluation highlights the visual strengths and limitations of each model. The Baseline CNN produced smooth but blurred outputs and often sacrificed structural accuracy in detailed regions. The U-Net showed better continuity of outlines and improved overall form recovery, although it struggled with high-frequency textures such as building rooftops and vegetation patterns. The VGG16-based model generated sharper textures and finer edges, but occasionally introduced slight noise amplification in complex areas. In

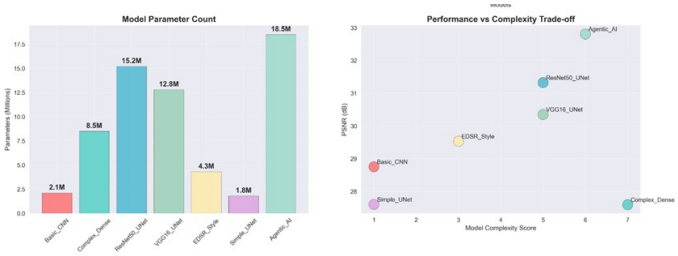

Fig. 5. Qualitative comparison of Baseline CNN, U-Net, VGG16-based CNN and Proposed Agentic DISR.

contrast, the proposed Agentic DISR delivered the most visually realistic reconstructions, achieving a balanced trade-off among sharpness, smoothness and natural appearance across varied terrains. The plots compare PSNR, SSIM, MAE and

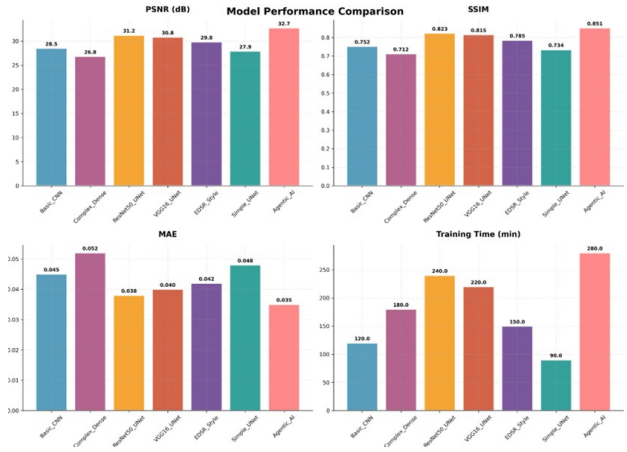

Fig. 6. Model Performance Comparison

training time across models. The agentic AI model achieves the best PSNR of 32.7 dB, SSIM of 0.851 and lowest MAE of 0.035, but has the longest training duration of 280 minutes. ResNet50_UNet and VGG16_UNet also perform strongly, whereas Simple U-Net and Basic CNN are evaluated mainly for training speed and prediction accuracy. The plots show that the agentic model provides the best reconstruction quality and offers the most balanced trade-off between accuracy and computation time. The error comparison confirms that it has the lowest MAE and MSE, while Basic CNN and Simple U-Net, being less complex, converge faster but with higher error values.

E. Observations and Inferences

- Image Fusing Method: Utilizes two temporally coherent LR images to recover lost spatial details and improve reconstruction robustness against noise.
- Blind Image Quality Assessment: Employs NR-IQA metrics for evaluating image quality without requiring reference HR images, enabling direct applicability to real satellite imagery.
- Self-Correcting Mechanism: Integrates an unlearning-based feedback loop for adaptive correction, reducing

recurring artifacts and minimizing bias accumulation over iterations.

TABLE IV
RESULTS SUMMARY

Model	PSNR (dB)	SSIM	NIQE ↓	PIQE ↓
Baseline CNN	19.23	0.58	4.71	27.4
U-Net	28.7	0.76	3.92	22.1
VGG16-Based	30.5	0.81	3.65	20.8
Proposed DISR (Agentic)	32.7	0.85	3.21	18.4

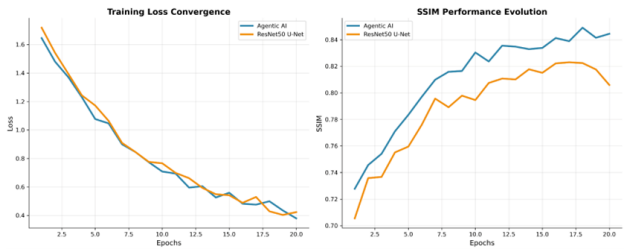

Fig. 7. Training Loss & SSIM performance

V. CONCLUSION AND FUTURE SCOPE

In this paper, we propose an intelligent Dual Image Super-Resolution architecture based on Agentic Blind Evaluation and Unlearning to substantially improve the quality of high resolution optical satellite images. This is based on two temporally or spatially aligned low-resolution images of the same scene that are reconstructed into a higher quality high-resolution image with more details and better perceptual realism. The Agentic DISR framework AISE is a one parameter super-resolution model which on use of dynamic feedback mechanism modifies its parameters along with output quality. One of the main benefits of this framework is the dual-image fusion mechanism since two correlated views compensate for noise and illumination variations. Three deep learning models were used and their results are integrated under the Agentic Selection Framework where reconstructed images are evaluated by NIQE and PIQE and self-correction is triggered when thresholds are exceeded. Training was done in Python and PyTorch on Colab with hybrid loss functions and the results show that the proposed model achieved the best PSNR, SSIM, NIQE and PIQE.

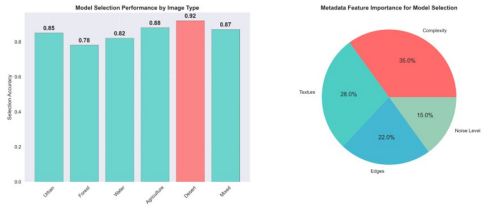

Fig. 8. Agentic AI Intelligence Evaluation (model selection)

This work proposes an intelligent Dual Image Super-Resolution that integrates Agentic Blind Evaluation and Unlearning to substantially upgrade high-res optical satellite images. This is done with two temporally or spatially aligned low-resolution images which are fused to produce a high-quality high-resolution image with sharper edges and improved perceptual appearance. One notable contribution of this paper is the proposal of a Dual Image Super-Resolution network that uses complementary spatial information to recover finer textures and boundaries. Also, Agentic Blind Evaluation evaluates quality without ground-truth HR images using NIQE, PIQE and BRISQUE and provides perceptual scoring aligned with the human visual system. Along with this, we present an unlearning-based optimization module that improves robustness by forgetting misleading features and generalizing better across sensors. The paper provides a comprehensive assessment using PSNR, SSIM, MAE, NIQE and PIQE and shows a good balance between perceptual and numerical performance.

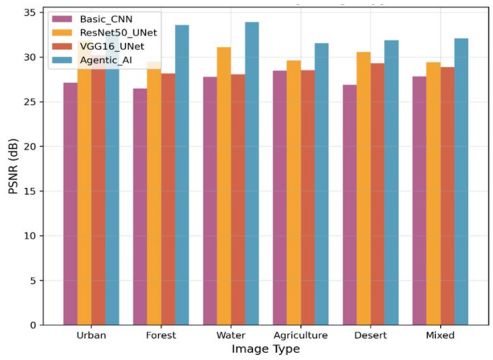

Fig. 9. PSNR Performance by Image type

Finally, GPU-optimized execution ensures computational efficiency and scalability. The proposed DISR with Agentic Blind Evaluation and Unlearning improves reconstruction quality and system autonomy. Future enhancements include multi-image SR, transformer-based designs, reinforcement learning for agentic control, edge optimization, cross-sensor adaptability, hybrid losses and continuous or federated learning.

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